Confined Space Rescue Plan

Prepared by:
Orange County Fire Chiefs' Association
Orange County Fire Services
Operational Plan

I. INTENT

To establish a method of operating for all Orange County Fire agencies at the scene of a Permit Required Confined Space (PRCS) incident.

II. PROCEDURE

This procedure is designed to identify standards for terminology, tactics, and support to be adopted by all participating agencies. All agency specific confined space policies and response plans shall comply with all applicable Cal-OSHA regulations (i.e. 5157, 5144, 3314, etc.) and should comply with applicable NFPA guidelines (i.e. 1670, 1006). This procedure is for use on incidents within the Orange County Operational Area.

A. Definitions

1. Confined Space
   a. Is large enough and so configured that an employee can bodily enter and perform assigned work; **and**
   b. Has limited or restricted means for entry and exit; **and**
   c. Is not designed for continuous employee occupancy.

2. Permit Required Confined Space - means a confined space that has one or more of the following characteristics:
   a. Contains or has a potential to contain a hazardous atmosphere; **or**
   b. Contains a material that has the potential for engulfing an entrant; **or**
   c. Has an internal configuration such that an entrant could be trapped or asphyxiated by inwardly converging walls or by a floor which slopes downward and tapers to a smaller cross-section; **or**
   d. Contains any other recognized serious safety or health hazard.

3. Hazardous atmosphere – an atmosphere that may expose employees to the risk of death, incapacitation, impairment of ability to self-rescue, injury, or acute illness from one of more of the following causes:
   a. Flammable gas, vapor or mist in excess of 10% of its lower flammable limit;
   b. Airborne combustible dust at a concentration that meets or exceeds its lower flammable limit;
      Note: This concentration can be approximated as a condition in which the dust obscures vision at a distance of 5 feet or less.
   c. Atmospheric oxygen concentration below 19.5% or above 23.5%.

Note: an atmosphere containing less than 19.5% oxygen is considered IDLH.

d. Atmospheric concentration of any substance for which a dose is published in Group 14 for Radiation and Radioactivity **or** a permissible exposure limit is published in Section 5155 for Airborne contaminants **and** which could result in employee exposure in excess of its dose or permissible exposure limit;

e. Any other condition that is immediately dangerous to life and health.
Note: Exceeding these values does not deny entry. They do, however, require the use of PPE and engineering controls as well as a detailed risk vs. benefit assessment.

4. Engulfment – the surrounding and capture of an Entrant by a liquid or finely divided flowable solid.

5. Entry – the action by which a person passes through an opening into a Permit-Required Confined Space. Entry is considered to have occurred as soon as any part of the Entrant's body breaks the plane of an opening into a Permit-Required Confined Space.

6. Immediately Dangerous to Life and Health (IDLH) – any condition that poses an immediate or delayed threat to life **or** that would cause irreversible adverse health effects **or** that would interfere with an individual's ability to escape unaided from a permit space.
Note: CCR 5144 Subsection (d) (1) (c) Where the employer cannot identify or reasonably estimate the employee exposure, the employer shall consider the atmosphere to be IDLH

7. Isolation – the process by which a permit space is completely protected against the release of energy and material into the space by such means as:
   - Blanking or blinding
   - Misaligning or removing sections of line, pipes, or ducts
   - Double block and bleed
   - Lockout/tagout
   - Blocking or disconnecting all mechanical linkages

   Note: Isolation should be completed by facility personnel whenever possible and confirmed and locked by Fire rescue personnel.

8. Permit Required Confined Space Rescue Team – as defined by the FIRESCOPE Field Operations Guide (ICS US&R-120-1), a minimum of

6 individuals trained to the Technician Level and equipped for Permit Space Entry and Rescue.

9. Entry Supervisor – the individual responsible for determining if acceptable entry conditions are present, for authorizing entry and overseeing entry operations, and for terminating entry. This individual is known as a Rescue Group Supervisor in ICS terminology and must be certified in Confined Space Entry and Rescue.

10. Rescue Entrant –trained rescue personnel assigned to perform the task of entry during rescue operations.

11. Rescue Attendant -trained rescue personnel assigned to perform the task of attendant during rescue operations involving entry-type rescue.

12. Assistant Safety Officer-US&R – as defined by FIRESCOPE, an individual who possesses the appropriate training to coordinate safety related activities for US&R operations. The Assistant Safety Officer – US&R reports directly to the Incident Safety Officer.

B. Initial Response Dispatch

As outlined in the *US&R and Technical Rescue Procedures* the following shall be considered a minimum initial response to a Confined Space Rescue in the Orange County Operational Area;

- 1 Battalion Chief
- 2 US&R Rescue Companies
- 1 Engine
- 1 Medic (may be combined with Engine if ALS Engine)
- 1 Truck
- 1 Haz Mat company
- 1 Light/Air
- 1 Ambulance

Agencies may choose to augment the minimum response with additional companies, specialized resources, Technical Experts, and/or overhead personnel as required.

C. Operational Goals non-US&R companies

The first arriving company shall attempt to locate a responsible party, obtain the initial entry permit, and determine the following;

- Number of victims
- Last known location

- Reason for initial entry
- Reason for rescue response
- Type of space

First due company should also establish a perimeter and keep all untrained persons from attempting a rescue.

First due Company Officer shall ensure adequate resources are assigned to the incident.

D. Operational Goals US&R Companies

1. The Company Officer of the first due US&R Company shall assume the role of *Rescue Group Supervisor* and assign the following key positions;
   a. Atmospheric Monitoring
   b. Rescue Attendant
   c. Rescue Entrant(s)
   d. Back Up Rescue Entrant(s)
   e. Rigging
   f. Ventilation
   g. Assistant Safety Officer US&R
   h. Breathing Air Systems (does not need to be an individual certified in confined space)

   Note: Key positions shall be filled by individuals holding certification in Confined Space Entry and Rescue. Some incidents may not require all key positions to be filled, nor is this key position list all inclusive. The incident may dictate special positions be created and filled.

2. The following guidelines should always be followed:
   a. Ensure a safe approach assessment to the space. This means monitoring the outside area from a known safe point and progressing towards the entry portal to determine the safety of the outside atmosphere where rescuers will be working (this may have to be completed while wearing SCBA).
   b. Obtain initial atmospheric readings of the PRCS. Ensure these readings are documented on the Confined Space Rescue Permit. Verify that atmospheric readings do not exceed permissible entry limitations for available PPE.
   c. Entrants shall continue to monitor the internal atmosphere in the area in which entry operations are taking place, unless this can be adequately accomplished from outside the space.

d. If ventilation is required, base type of ventilation on the properties of the atmospheric hazard (do not use gasoline powered ventilators).
e. Ensure space isolation has been completed and appropriate Agency approved rescuer locks/tags have been affixed to all energy isolation points. Keys to locks should be given to the incident Safety Officer or Assistant Safety Officer – US&R.
f. Follow Cal-OSHA requirements for back up rescue teams.
g. Atmosphere supplying respirators (SCBA/SAR) must be worn when entering a PRCS that contains, or may contain, a hazardous atmosphere. SCBA shall never be removed to enter, exit, or work within a space.
h. Nomex outer garments (or other Agency or NFPA approved Technical Rescuer clothing) should be worn when entering a PRCS.
i. All entrants shall wear an approved Class III harness with a retrieval line attached at a suitable point so that when retrieved, the entrant presents in the smallest possible profile unless this practice creates a greater overall risk of entry or would not contribute to the rescue of the entrant(s).
j. Retrieval lines shall be attached to a mechanical device or fixed point. Operations in vertical type permit spaces greater than 5 feet deep require that a mechanical device (approved man rated winch or rope system) be available to retrieve entrants and victims.
k. Ensure that all entries into a PRCS comply with, to the extent feasible, all applicable Cal-OSHA regulations.
l. The Rescue Group Supervisor shall ensure that a Confined Space Rescue Permit is completed and signed prior to entry. (See attachment)
m. The completed Confined Space Rescue Permit shall be retained by the Agency with jurisdictional authority for the PRCS for a minimum of one year.

E. Post Incident
1. Ensure all personnel and equipment is accounted for.
2. Ensure Confined Space Rescue Permit is cancelled.
3. In accordance with CCR Title 8 Section 342 subsection (b) immediately notify Cal-OSHA and provide the following information:
   a. Time and date of accident.
   b. Employer's name, address and telephone number.
   c. Name and job title, or badge number of person reporting the accident.
   d. Address of site of accident or event.

e. Name of person to contact at site of accident.
f. Name and address of injured employee(s).
g. Nature of injury.
h. Location where injured employee(s) was (were) moved to.
i. List and identity of other law enforcement agencies present at the site of accident.
j. Description of accident and whether the accident scene or instrumentality has been altered.

4. The Agency with jurisdictional authority of the rescue/entry shall retain the permit for a minimum of one year.

F. Training

1. Each Agency is responsible for ensuring that their certified personnel practice making Permit Space rescues *at least once* every 12 months.
2. This annual refresher training can be completed at actual PRCS's or representative spaces.
3. Training shall ensure rescuer proficiency as well as introduce new or revised procedures, equipment, etc.
4. Each Agency is responsible for maintaining sign in sheets and/or training records to serve as proof of currency of its rescuers.

CONFINED SPACE ASSESSMENT GUIDE

A. <u>Is the space in question:</u>

1. Large enough and so configured that a person can bodily enter? ☐ Yes ☐ No
2. Have limited or restricted means of entry or exit? ☐ Yes ☐ No
3. Not designed for employee occupancy? ☐ Yes ☐ No

If you answered <u>YES</u> to questions 1-3, the space in question is a <u>CONFINED Space</u> as defined by Cal-OSHA. Proceed to questions 4 -7 to determine if the confined space is PERMIT REQUIRED.

B. <u>Does the confined space in question have any one of the following characteristics?</u>

1. A potential atmospheric hazard? ☐ Yes ☐ No
   (Oxygen deficiency or enrichment/Flammability/Toxic or unknown)
2. A potential engulfment hazard (liquid or flowable solid)? ☐ Yes ☐ No
3. An internal configuration hazard (inwardly converging walls)? ☐ Yes ☐ No

1. Any other recognizable safety or health hazards (Noise, moving parts of machinery, etc.) ☐ Yes ☐ No

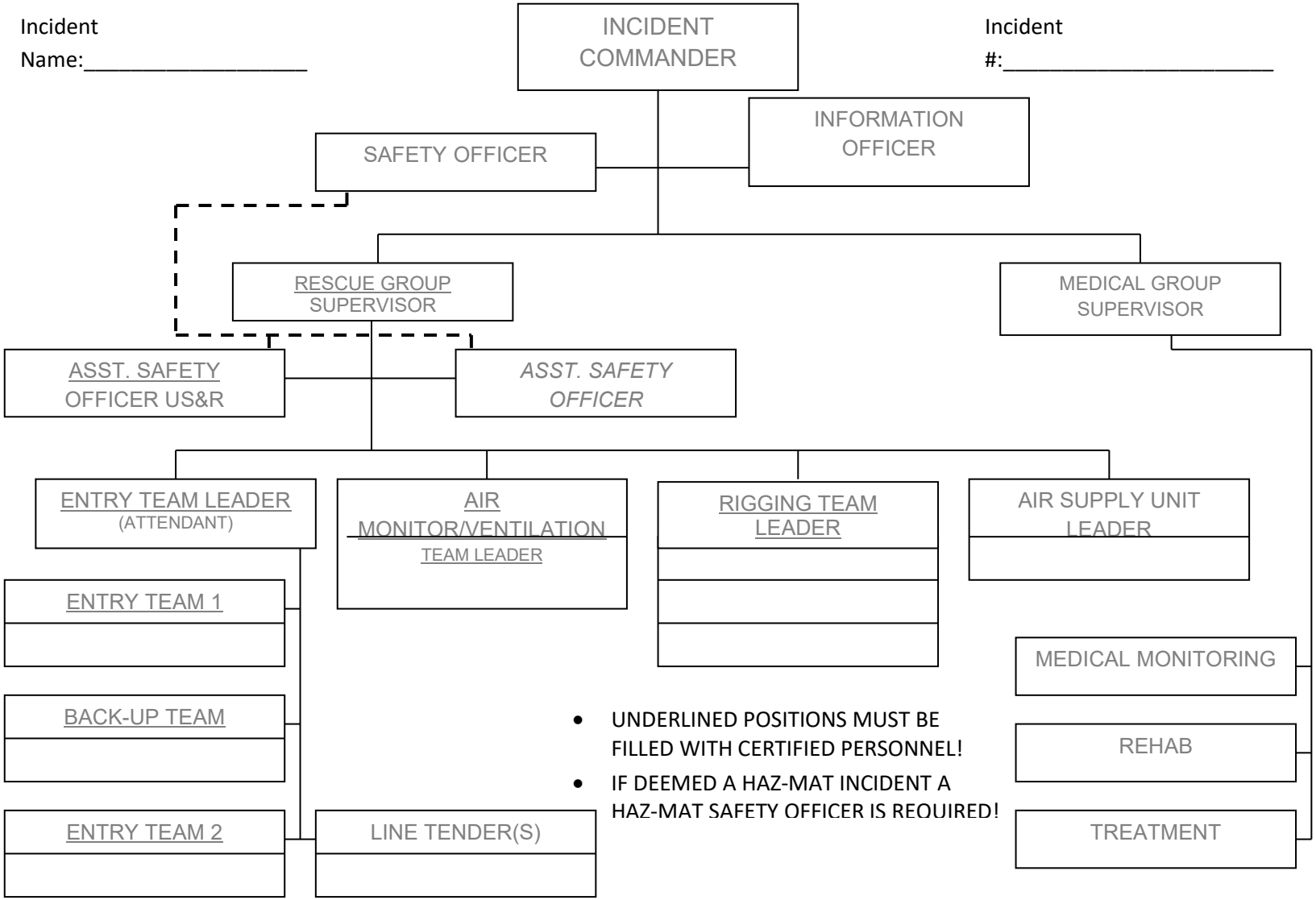
If you answered <u>YES</u> to any of the questions in 4-7 the space is a PERMIT REQUIRED CONFINED SPACE as defined by Cal-OSHA.

- Initiate Confined Space Rescue Permit and Procedures.
- Isolate and Deny Further Entry.
- Attempt to Retain a Responsible Person.
- Request Resources - Notify the appropriate US&R Companies/Crews, 2nd Alarm, Light/Air, Haz Mat (if necessary).

CONFINED SPACE RESCUE ORGANIZATIONAL CHART

Incident Name:_____

Incident #:_____



- UNDERLINED POSITIONS MUST BE FILLED WITH CERTIFIED PERSONNEL!
- IF DEEMED A HAZ-MAT INCIDENT A HAZ-MAT SAFETY OFFICER IS REQUIRED!

ORANGE COUNTY FIRE SERVICES - CONFINED SPACE RESCUE PERMIT

SCENE ASSESSMENT

Incident Date: _____ On-scene Time: _____ Incident #: _____

Incident Location: _____

Space I.D# (if applicable) _____Contents of Space_____

Initial Entry Permit Available? Yes No / M.S.D.S. On-Site? Yes No

Reason for Original Entry: _____

# of Victims _____ History of Accident:_____

Last Contact With Victims (Time and Location)_____

Was On-Site Rescue Attempted? Yes No

If Yes, Why was Rescue Not Completed? _____

Responsible Party / Site Manager /Owner _____

PRE-ENTRY OPERATIONS

HAZARD IDENTIFICATION

ATMOSPHERIC — Bump Test Performed – Yes No

Oxygen _____% <19.5% IDLH >23.5% = Permit Required

LEL _____% >10% LEL = Permit Required

Carbon Monoxide (CO) _____ppm >25 ppm TLV= Permit Required (1200 ppm IDLH)

Hydrogen Sulfide (H2S) _____ppm >10 ppm PEL= Permit Required (100 ppm IDLH)

Other _____ppm of _____

Time Tested_____ Person Testing_____ Calibration Date_____

ELECTRICAL ☐ _____
PNEUMATIC ☐ _____
HYDRAULIC ☐ _____
MECHANICAL ☐ _____
OTHER ☐ _____

ESTABLISH RESCUE OBJECTIVE: RESCUE ☐ BODY RECOVERY ☐

HAZARD CONTROL

VENTILATION	☐ Positive Pressure	☐ Negative Pressure	☐ Combination
ELECTRICAL	☐ Lockout	☐ Tagout	
PNEUMATIC	☐ Lockout	☐ Tagout	
HYDRAULIC	☐ Lockout	☐ Tagout ☐ Bleed Lines	☐ Disconnect Lines
MECHANICAL	☐ Block linkage	☐ Disconnect	
PIPING	☐ Lockout	☐ Tagout ☐ Bleed Lines	☐ Disconnect Lines
OTHER	☐ (Specify Method) _____		

RESOURCES AND EQUIPMENT REQUIRED

PRE-ENTRY OPERATIONS

Resource Request: ☐ US&R Teams ☐ Light/Air ☐ Haz. Mat.
☐ Special Equip. (air knife, vac truck, lumber, etc.)

Resp. Protection: ☐ SCBA ☐ SAR

PPE: ☐ NFPA Approved Clothing ☐ Helmet ☐ Gloves
☐ Tyvek Suits ☐ Class 3 Harness

Gas Monitors: ☐ _____

Lighting: ☐ Headlamp ☐ Hand light ☐ Light Sticks
☐ Cord Light (explosion proof/GFCI, if required)

Entry/Extrication: ☐ High Point Anchor _____ (type) ☐ Class 3 Harness
☐ Rope / Winch MA System ☐ Belay Line (Tag Line)

Victim Package: ☐ Backboard ☐ Half Back ☐ SKED
☐ Stokes Litter ☐ Harness ☐ Other _____

Communications: ☐ Visual ☐ Hardwire (Con-Space) ☐ Radio

ENTRY OPERATIONS

Pre Entry Briefing Accomplished - ☐ @ _____ Hours
Rescue Group Supervisor: _____
Entry Team Leader / Rescue Attendant: _____
Entry Team # 1: _____
Back Up Team / RIC: _____
Entry Team # 2: (if Req'd)_____
Assistant Safety Officer US&R_____
Rescue Group Supervisor: (Print)_____
(Signature)_____ Entry Authorized ☐

SITE SAFETY PLAN

SITE SAFETY PLAN

N ↑

TERMINATION

Entry Terminated: Time:_____ Date: _____

Rescue Group Supervisor: _____

(Signature:) _____

ATMOSPHERIC MONITORING LOG

Incident #	Incident Name:
Date:	Time:

Continuous Monitoring – Document Readings Approximately Every _____ Minutes

TIME	**(O_2) OXYGEN** < 19.5- IDLH or > 23.5 %	**(LEL) FLAMMABLE** >10% LEL – Permit Required	**(CO) CARBON MONOXIDE** 25 ppm -TLV/ 1200 PPM-IDLH	**(H_2S) HYDROGEN SULFIDE** 10 ppm – PEL / 100 ppm - IDLH	**OTHER**

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TIME	**(O_2) OXYGEN** < 19.5- IDLH or > 23.5 %	**(LEL) FLAMMABLE** >10% LEL – Permit Required	**(CO) CARBON MONOXIDE** 25 ppm -TLV/ 1200 PPM-IDLH	**(H_2S) HYDROGEN SULFIDE** 10 ppm – PEL / 100 ppm - IDLH	**OTHER**

BREATHING AIR SUPPLY MONITORING LOG

Incident #	Incident Name:
Date:	Time:

"Continuous Monitoring Required"

TIME	PRESSURE (psi)	ENTRANTS NAME	HOSE ID	AIR SOURCE SCBA, SABA, 30 MIN/1 HR, CART, OR COMPRESSOR	AIR RESUPPLY AVAILABLE # _____	NOTES

TIME	PRESSURE (psi)	ENTRANTS NAME	HOSE ID	AIR SOURCE SCBA, SABA, 30 MIN/1 HR, CART, OR COMPRESSOR	AIR RESUPPLY AVAILABLE # _____	NOTES

CONFINED SPACE RESCUE TACTICAL LOG

Incident #	Incident Name:
Date:	Time:

TIME	LOG ENTRY

TIME	LOG ENTRY